GrapeMaster: A Digital Agriculture Platform for Crop-Season-Centered Closed-Loop Vineyard Management

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Abstract

Grapevine is a high-value fruit crop, but downy mildew and other epidemic diseases remain major constraints on yield, fruit quality, plant-protection costs, and sustainable production in humid subtropical regions. Vineyard plant-protection management is inherently a crop-season-scale decision process that should link phenological progression, weather-driven risk, season-long agronomic advisory, field execution, fungicide protection, symptom evidence, and record review. However, the difficulty of observing pre-symptomatic infection risk, together with experience-based operations, missing execution feedback, and fragmented records, prevents point capabilities such as disease warning, phenology simulation, image recognition, and advisory tools from naturally becoming a continuous management loop. Here, we designed, developed, deployed, and workflow-verified GrapeMaster, a crop-season-centered orchard management platform that uses a unified crop-season data model and a risk-task-feedback-evidence-archiving workflow to transform trustworthy analytical outputs into executable, feedback-enabled, and traceable management actions. This workflow is supported by weather-driven grape phenology simulation, pre-symptomatic downy mildew first-infection warning, grape disease image recognition, and bounded advisory interpretation. Backend operational records, module-validation evidence, and representative crop-season replay showed that GrapeMaster can transform disease-risk information into management tasks, protection states, subsequent risk updates, and traceable records. This study provides a deployable platform framework for shifting vineyard management from fragmented experience-based decisions toward crop-season-centered, evidence-driven, and traceable closed-loop management, supporting future eval-

uations of pesticide-use efficiency, regional services, and industry-scale application.

Keywords: grapevine; downy mildew; decision support system; crop-season management; digital agriculture; large language model; closed-loop management

1 Introduction

Grapevine is a high-value fruit crop for fresh consumption, wine production, and processing, and disease management directly affects yield stability, fruit quality, plant-protection inputs, and sustainable production. Humid subtropical vineyards face frequent high humidity, rainfall, dew formation, rapid canopy growth, and heterogeneous field microclimates, all of which favor foliar diseases such as downy mildew, powdery mildew, gray mold, and anthracnose (GESSLER et al., 2011; Boso et al., 2014; Toffolatti et al., 2024). Downy mildew is a particularly representative constraint because its early infection risk is governed by weather, host phenology, inoculum pressure, and local microenvironment before visible symptoms can be observed. This makes early-season risk difficult to judge from field inspection alone and helps explain why grape disease control often relies on experience-based or calendar-based interventions under high disease pressure (Pertot et al., 2017; Savary et al., 2018; Tudi et al., 2021; Oliver et al., 2023).

The pre-symptomatic first-infection risk of downy mildew is not only a single-disease modelling problem; it also illustrates a broader management break in crop-season-scale vineyard protection. Risk may develop before symptoms are visible, whereas management responses must account for phenological timing, weather-driven risk, season-long agronomic advisory, task scheduling, fungicide protection, symptom diagnosis, and record review. Vineyard plant protection is therefore not a single warning, diagnosis, or recommendation. It is a continuous decision process across the crop season. In practice, this process is often fragmented by experience-based decisions, fixed-interval operations, manual reminders, paper records, messaging-app records, or single-purpose software functions. These arrangements can leave risk assessment, recommendation, field execution, protection feedback, symptom evidence, and retrospective review without a shared data object or workflow constraint.

Weather-driven disease models and DSSs provide a strong scientific basis for risk-based plant protection. In grapevine downy mildew, models have progressed from empirical warning rules to mechanistic or semi-mechanistic descriptions of oospore maturation, infection, incubation, sporulation, and epidemic development (Orlandini et al., 1993; Park et al., 1997; Caffi et al., 2011b). Warning systems have shown potential to improve treatment timing when meteorological inputs, local calibration, and implementation context are adequate (Caffi et al., 2010; CAFFI and ROSSI, 2018; Carisse, 2016). Grapevine phenology models likewise provide crop-stage information needed to interpret host susceptibility, warning windows, canopy development, and operation timing (Caffarra and Eccel, 2010; Parker et al., 2011; García de Cortázar-Atauri et al., 2017; Wang et al., 2020). Image-recognition models can convert visible symptoms into structured diagnostic evidence, and recent vision-language or LLM-based agricultural systems have begun to connect perception, explanation, and recommendation (Yan et al., 2024, 2025; Shafay et al., 2026; Rai et al., 2026). These technologies are valuable, but they frequently remain point outputs: a risk level, a phenological stage, a disease class, or an advisory text. They can support crop-season management only when they enter an executable, feedback-enabled, and traceable workflow.

Digital agriculture and DSS implementation studies increasingly show that model performance is only one part of production deployment. Operational systems also need stable data ingestion, reliable service operation, modular architecture, user-facing task processes, feedback mechanisms, record tracking, and workflow verification. End-to-end IoT and remote-sensing platforms demonstrate how heterogeneous data streams, analytics, and interfaces can be connected for complex agricultural management (Bakir et al., 2026; Nkwocha and Chandel, 2025). Public DSS and implementation studies further show that decision tools must be accessible, interpretable, and compatible with actual advisory and grower workflows (Seem, 2004; Rossi et al., 2014; Bregaglio et al., 2022; Rossi et al., 2023; González-Domínguez et al., 2023). These findings shift the research problem from whether an individual model can predict a biological event to whether trusted analytical capabilities can be operationalized as a stable management system.

For grape production, this platform-level problem remains insufficiently resolved. Disease-risk models can identify infection windows, but warnings often stop before task generation and execution feedback. Phenology models can predict crop-stage progression, but their outputs are not always connected to risk interpretation, advisory timing, or management records. Image-recognition tools can identify symptomatic leaves, but they are rarely linked with pre-symptomatic risk forecasts and crop-season history. LLM advisory functions can improve explanation and interaction, but they may be detached from field, crop-season, risk, task, pesticide-use, and record contexts (Chen et al., 2026; Puelles et al., 2024). Grapevine management therefore lacks a crop-season-centered platform that organizes pre-symptomatic risk warning, task execution, fungicide-protection feedback, dynamic risk updating, post-symptomatic recognition, advisory interpretation, and record archiving into the same closed-loop process.

To address this gap, we designed, developed, deployed, and workflow-verified GrapeMaster, a crop-season-centered digital platform for vineyard management. GrapeMaster uses the crop season as the core management object and organizes phenology simulation, weather-driven downy mildew risk warning, agronomic advisory, plant-protection task generation, fungicide-protection feedback, symptom-image recognition, bounded LLM consultation, and record archiving within one workflow. The platform is supported by a downy mildew first-infection risk engine calibrated for humid subtropical vineyards, a weather-driven grape phenology module, a grape-focused disease-recognition module based on retrained visual modelling, and an advisory module derived from a published crop-disease vision–language framework. The objective of this paper is to present GrapeMaster as a deployable platform-system framework and to evaluate whether its data model, analytical modules, backend records, and representative crop-season replay can transform disease-risk information into executable, feedback-enabled, and traceable vineyard management actions.

2 Materials and methods

2.1 Study design, evidence sources, and vineyard context

2.1.1 Study design and verification strategy

This study describes the design, development, deployment, and workflow verification of GrapeMaster, a self-developed system platform for crop-season-centered grapevine plant-protection management. The platform was organized to support crop-season setup, weather and phenology updating, disease-risk warning, plant-protection task management, post-treatment protection-state updating, image-assisted disease diagnosis, LLM-assisted consultation, and management-record archiving. The study object is a single GrapeMaster platform. Platform implementation records are used to verify whether the mobile application, backend services, database objects, scheduled analytical workflows, notification and task services, image-upload pathway, LLM consultation entry, and crop-season record system can operate as an integrated workflow. Model validation and module-specific evidence are used to support the analytical components embedded in that workflow.

In the current publication configuration of GrapeMaster, the analytical layer includes a downy mildew initial-infection model for pre-symptomatic risk warning, a phenology module for crop-stage interpretation, and grape-focused image-recognition and LLM advisory modules for post-symptomatic evidence and bounded consultation. These modules are evaluated according to their roles in the platform rather than as independent standalone systems.

2.1.2 Study area and field data sources

The analytical modules were configured and evaluated for humid subtropical vineyard management using field data from Guangxi, China. This evidence base included meteorological records, grapevine phenological observations, airborne sporangia monitoring, symptom observations, and management records collected from field experiments and related vineyard observations. The same regional data context supports the FSIM-S downy mildew risk engine, the Guangxi phenology calibration, and part of the grape disease image expansion used for platform-specific recognition.

This regional evidence is used to support the current GrapeMaster configuration and workflow verification in humid subtropical vineyards. It is not treated as unconditional cross-regional validation. Extension of the platform to other grape-growing regions would require local weather data, phenological calibration, disease-risk parameterization, and management-rule adaptation.

2.2 Platform architecture and crop-season data model

2.2.1 Software architecture and service layers

GrapeMaster uses a layered architecture in which the mobile client, backend business services, analytical model services, persistent storage, and external data services are separated but connected through REST APIs, scheduled jobs, WebSocket channels, and database records. The central design unit is the crop season, which links a vineyard field with cultivar information, cultivation method, vine age, weather history, phenology outputs, disease-risk simulations, fungicide applications, plant-protection tasks, notes, image records, messages, and notifications.

The platform was implemented with a mixed programming stack selected according to layer responsibility. The mobile application was written in Dart with Flutter, with an Android shell configured through Kotlin and Gradle. The backend business service and scheduled analytical workflows were implemented in Python using Django 4.2, Django REST Framework, Django Channels, Django APScheduler, SimpleJWT, and PostgreSQL access through `psycopg2`. The disease-model service was implemented as an independent Python service exposed through FastAPI, Uvicorn, Pydantic, and Starlette. Numerical and time-series processing used NumPy, Pandas, Python Dateutil, and Pytz. Geospatial and weather-related processing used packages including Shapely, Fiona, GeoPandas, Geopy, Requests, HTTPX, Open-Meteo client packages, request caching, and retry utilities. Image-processing and recognition workflows used Python deep-learning and image-processing libraries, including PyTorch, TorchVision, OpenCV, Pillow, Ultralytics components, EfficientViT-related modules, and segmentation components.

The logical architecture was organized into five layers (Table 1), covering application interaction, data ingestion, analytical modeling, image-based diagnosis, alerting, and operational feedback.

Table 1. Logical architecture layers of GrapeMaster used in the current study.

Layer	Main components	Main responsibilities
Application and presentation layer	Dart/Flutter mobile modules for field, crop-season, task, recognition, map, note, authentication, weather, and notification functions	Field input, weather and warning display, plant-protection task operation, image upload, consultation entry, and grower-facing interaction
Data ingestion and storage layer	Django REST Framework, PostgreSQL, Django Channels, Django APScheduler, SimpleJWT, master-data tables, and backend business objects	API routing, authentication, data persistence, scheduled updates, WebSocket routing, field records, crop-season records, and operational records
Weather, phenology, and disease analytics layer	Weather preprocessing, Pandas, NumPy, phenology workflow, spraying-suitability workflow, and FastAPI disease-model interface	Weather transformation, crop-stage simulation, disease-risk request construction, treatment-window generation, and risk interpretation
Image recognition and LLM advisory layer	Image upload endpoints, PyTorch-based image models, segmentation-related processing, consultation interface, and advisory-response records	Symptom-image processing, disease-category output, visual evidence storage, LLM-assisted explanation, and post-symptomatic advisory support
Task recommendation, alert, and feedback layer	Notice services, task services, fungicide and pesticide records, treatment windows, and protection-state feedback	Warning delivery, task creation, task completion, overdue handling, fungicide-history conversion, and risk recalculation after completed operations

2.2.2 *Field and crop-season records*

The data model was organized around two primary agricultural entities: the vineyard field and the crop season. The field record stores the user, field name, area, GeoJSON boundary, centroid longitude and latitude, administrative region, sharing state, deletion state, notice state, and generated field-boundary preview. When a field is saved, the backend parses the GeoJSON boundary, derives a boundary image, estimates field area, calculates the centroid, and requests geocoding information where needed.

The crop-season record links the field with grape cultivar, cultivation method, vine age, field coordinates, field size, and cultivar metadata. It is the main relational anchor for weather records, phenology outputs, disease-risk outputs, fungicide and pesticide records, plant-protection tasks, irrigation tasks, notes, notifications, image records, and user-facing summaries. This design was selected because the same vineyard field persists across years, whereas disease risk, weather, phenology, protection history, observations, and management actions are season-specific.

Master-data tables were used as controlled reference tables for the two anchors rather than as independent management objects. Grape variety, cultivar susceptibility, cultivation method, BBCH stage, disease type, fungicide, pesticide, and reminder-rule tables provide standardized attributes that are selected when a field, crop season, plant-protection task, risk record, or advisory record is created. In this structure, the field and crop season remain the relational anchors, while the master tables supply reusable biological, agronomic, and operational descriptors. The implementation stores both normalized relational records and JSON model payloads so that weather, phenology, disease-risk, and model request/response records can be displayed to the mobile client while preserving the original inputs and outputs needed for recalculation, auditing, and later validation.

2.3 Weather ingestion and crop-season preprocessing

For each active crop season, scheduled backend workflows retrieve historical and forecast weather using the field centroid. The weather workflow combines historical observations with forecast records where available. Hourly variables include air temperature, relative humidity, dew point, precipitation, wind speed, and wind gusts. Daily variables include maximum, minimum, and mean temperature, precipitation, maximum wind speed, maximum gust speed, sunrise time, and sunset time. Missing values are handled before analytical use so that downstream workflows receive complete date-aligned input series.

The backend transforms weather data into three working schemas. Daily mean temperature is converted into phenology-model input; hourly weather is converted into the disease-model schema, including datetime, dew point, precipitation, relative humidity, air temperature, and wind speed; and short-term hourly records are used to evaluate spraying suitability. This preprocessing belongs to the GrapeMaster platform workflow and supplies input data to the analytical modules operationalized in GrapeMaster.

2.4 Analytical modules operationalized in GrapeMaster

2.4.1 Phenology simulation and Guangxi calibration

The phenology workflow estimates grapevine growth-stage progression from daily temperature and crop-season metadata. In the current platform implementation, the backend calculates daily thermal accumulation and maps accumulated development to BBCH-related growth-stage classes for mobile display and disease-risk interpretation. The phenology output has two platform functions: it gives users a crop-development timeline and supplies disease-risk modules with a daily crop-stage sequence.

In GrapeMaster, the phenology module is defined as a platform submodule rather than as an independent model paper. The module provides crop-stage timing for disease-risk interpretation, warning windows, and management-task scheduling. The current publication configuration uses Guangxi phenological observations and platform weather inputs to support crop-season interpretation in humid subtropical vineyards.

2.4.2 FSIM-S downy mildew first-infection risk engine

In GrapeMaster, the current operational disease-risk service receives weather, crop-stage sequence, cultivar susceptibility, target disease information, and fungicide-history inputs through the platform interface and returns date-aligned disease-risk states, field-risk states, protection states, recommendations, and treatment windows. The current operational service is described as a team-developed disease-model module rather than as a simple empirical logic rule. Its role in this paper is to demonstrate that the platform can pass model inputs, store model outputs, display risk states, trigger warnings, and feed completed tasks back into later risk states.

In GrapeMaster, the disease-risk module is configured with FSIM-S, a first seasonal infection model developed for humid subtropical grape-growing areas in Guangxi. The module predicts the first seasonal infection of grapevine downy mildew under conditions where early-season infection may be driven by recurrent airborne asexual sporangia rather than by a discrete oospore-driven primary infection event. It was calibrated and validated using 2022–2024 site-year observations from Guangxi, including phenology, airborne sporangia, symptom onset, management records, and meteorological data.

FSIM-S uses a mechanistic infection chain adapted from Rossi primary-infection modelling and related sporangia-based infection modelling work (Rossi et al., 2008; Caffi et al., 2009, 2011a; Brischetto et al., 2021). The module represents sporangia production, accumulation, dispersal, deposition, zoospore release and survival, infection establishment, incubation, and symptom expression. In the subtropical setting, airborne sporangia are treated as the relevant inoculum source for the first seasonal infection. Rainfall-driven dispersal is represented as rapid deposition onto plant surfaces, whereas wind-driven dispersal allows airborne sporangia to decay before later deposition. The platform implementation uses field-derived spore pressure to regulate the infection-triggering threshold. A daily sporangia pressure field $S(d)$ is constructed from field spore monitoring data after interpolation, normalization, and smoothing. Hourly pressure is represented as a three-day rolling mean:

$$P_t = \frac{1}{3} \sum_{k=0}^{2} S(d_t - k), \tag{1}$$

where P_t is the spore-pressure index at hour t, and d_t is the corresponding calendar day. Infection is triggered when the wetness-duration term and its temperature response exceed a spore-pressure-adjusted effective threshold:

$$\begin{gathered} IC_t^* = \begin{cases} IC \exp(-\alpha_{\text{pos}} P_t), & P_t \geq 0, \\ IC \exp\left[\alpha_{\text{neg}}(-P_t)\right], & P_t < 0, \end{cases} \\ \text{infection}_t = \mathbb{I}\left(WD_t\, TWD_t \geq IC_t^*\right), \end{gathered} \tag{2}$$

where WD_t is the wetness-duration component, TWD_t is the temperature response during the wetness period, IC is the base infection threshold, α_{pos} and α_{neg} are calibration parameters for positive and negative spore-pressure effects, and $\mathbb{I}(\cdot)$ is an indicator function. Positive spore pressure lowers the effective infection threshold, whereas low pressure raises it.

The GrapeMaster disease-risk workflow focuses on downy mildew because FSIM-S is calibrated for that target disease. Other grape diseases may be supported by recognition or advisory functions, but they are not treated as equally validated risk-prediction targets in this study. Weather inputs can be supplied through public weather services, grid weather products, regional weather data, local weather stations, or other sources that can be transformed into the platform schema. The spore measurements used to develop FSIM-S are treated as model-development evidence and as the basis for parameterizing or reconstructing spore-pressure scenarios; they are not described as a live data stream required from every vineyard during routine platform operation. Detailed parameter definitions, sensitivity analysis, calibration settings, and validation results for FSIM-S are reserved for the model-specific supplementary material.

Figure 1 presents the platform-level operationalization of the disease-risk engine rather than the full mathematical specification of FSIM-S. It therefore includes both model-related transformations and user-facing outputs generated or displayed by the GrapeMaster workflow.

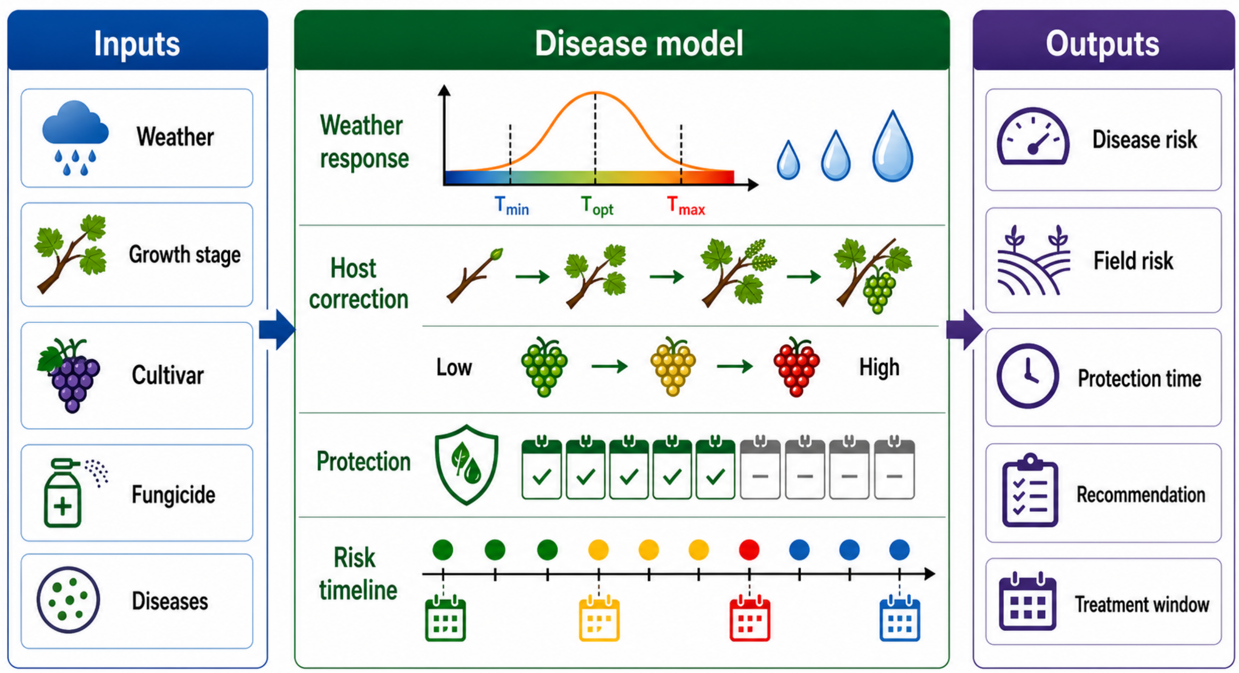


Figure 1. Platform-level operationalization of the GrapeMaster disease-risk engine. Weather, growth-stage, cultivar, fungicide-protection, and target-disease inputs are transformed through weather-response, host-correction, protection-state, and risk-timeline components into user-facing outputs, including disease-risk states, field-risk states, protection-time estimates, recommendations, and treatment windows. The figure summarizes how the disease-risk engine is embedded in the platform workflow and is not intended as a complete mathematical specification of FSIM-S.

2.4.3 Grape disease recognition and bounded LLM advisory

The image-based workflow supports symptom-level evidence after disease signs are observed or suspected. Users can select images from the camera or gallery, preview them in the mobile client, compress them before upload, and submit them through backend image-processing endpoints. The backend image-recognition service returns disease or healthy categories and associated diagnostic or management text for mobile display. Image outputs can be linked to crop-season notes, disease reports, and advisory records, allowing visual evidence to remain within the same field-season context as weather, phenology, risk, and task records.

In GrapeMaster, the image-recognition and advisory module follows the published CDIP-ChatGLM3 dual-model framework, which combines a deep-learning image classifier with a fine-tuned ChatGLM3-based advisory model for crop disease identification and prescription (Yan et al., 2025). The platform-specific update is restricted to the grape scenario. The grape disease image dataset is expanded with additional field photographs, and two grape disease categories are added to the original grape-relevant label set. The same candidate vision-model families, training workflow, validation procedure, and vision–LLM integration structure described in the reference framework are retained. The final server-side deployment model is selected from the retrained candidate models according to grape-focused validation performance.

For image recognition, each candidate model outputs class probabilities using a softmax layer and is trained with cross-entropy loss:

$$p(y = k \mid x) = \frac{\exp(z_k)}{\sum_{j=1}^{K} \exp(z_j)}, \qquad \mathcal{L}_{\mathrm{CE}} = -\sum_{k=1}^{K} y_k \log p(y = k \mid x), \tag{3}$$

where x is the input image, z_k is the logit for disease class k, K is the number of grape disease or healthy categories, and y_k is the one-hot class label. Model selection is based on grape-focused validation metrics, including accuracy, precision, recall, and F1-score:

$$m^* = \arg\max_{m \in \mathcal{M}} \mathrm{Score}_{\mathrm{val}}(m), \tag{4}$$

where $\mathcal{M}$ is the set of candidate vision models and $\mathrm{Score}_{\mathrm{val}}$ is the selected validation criterion, such as accuracy or macro-F1. The selected classifier provides the disease label and confidence score used by the advisory interface.

The LLM-assisted advisory component follows the CDIP-ChatGLM3 interaction structure: image-recognition outputs are converted into disease-specific consultation context, and the fine-tuned language model generates management-oriented advisory text. Advisory evaluation is restricted to grape disease consultation questions and uses domain-specific consultation cases together with text-generation metrics such as BLEU and ROUGE. The LLM output is treated as advisory text and knowledge support. It does not replace the disease-risk model, expert diagnosis, pesticide labels, or plant-protection regulations. In the platform workflow, LLM messages and advisory outputs are stored or linked to the crop-season context where possible so that consultation becomes part of the management record rather than an isolated chat. Figure 2 summarizes this bounded image-recognition and advisory workflow.

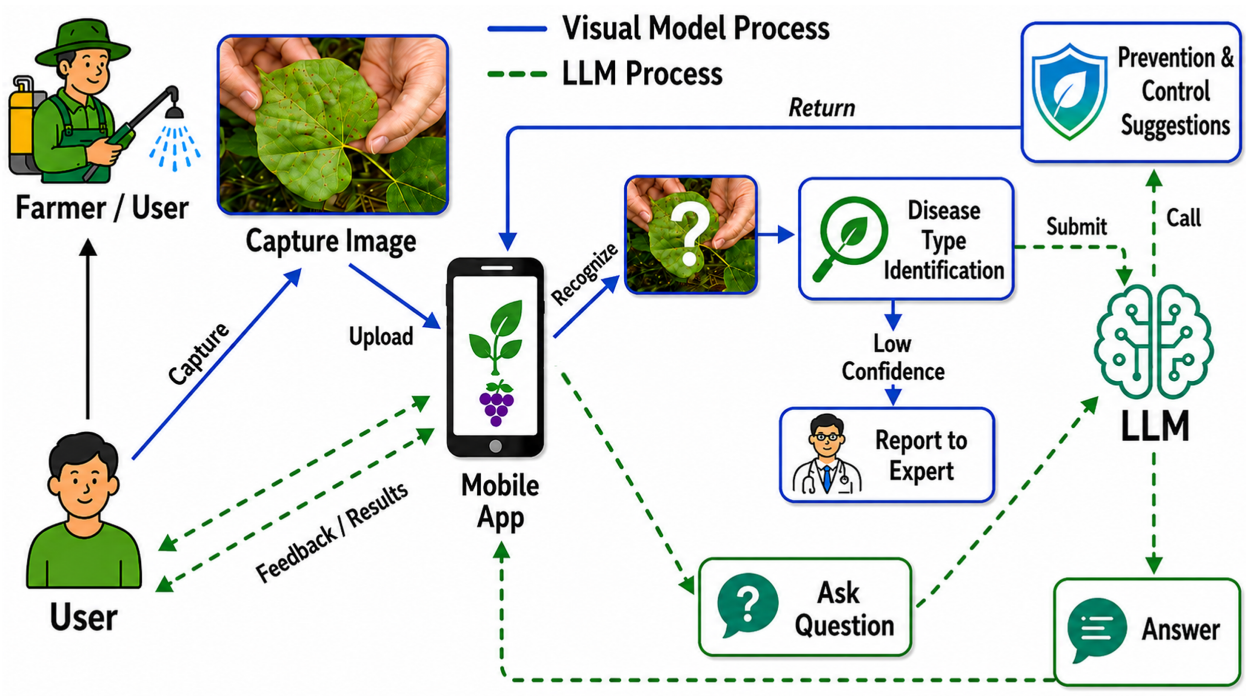


Figure 2. Image-recognition and bounded advisory workflow in GrapeMaster. A user captures and uploads a field image through the mobile application, after which the visual model provides a candidate grape disease category and confidence information. Low-confidence cases can be directed to expert review. The LLM pathway is used for auxiliary agronomic consultation and management-oriented explanation based on the recognition context; it is not positioned as a replacement for validated disease-risk models, expert diagnosis, pesticide labels, or local plant-protection regulations.

2.5 Module provenance and operationalization table

Table 2. Provenance and operational roles of the main GrapeMaster modules.

Module	Source or provenance	Platform role	Evidence and boundary
Platform workflow	GrapeMaster backend, mobile client, database, scheduled jobs, and notification/task services	Links field, crop season, weather, phenology, risk, task, fungicide, image, advisory, and record objects	Backend records verify workflow operability and data linkage, not user adoption or economic outcomes
Phenology module	GrapeMaster weather-driven phenology workflow, calibrated with Guangxi field observations	Provides crop-stage timing for risk interpretation, advisory timing, and task scheduling	Supports crop-season interpretation in the Guangxi humid subtropical context; broader use requires local calibration
FSIM-S risk engine	Team-developed downy mildew first-infection model calibrated and validated with Guangxi field data	Provides pre-symptomatic downy mildew risk windows for warning and task triggering	Validated for downy mildew first-infection risk; not a risk model for all grape diseases
Grape disease recognition	Grape-focused retraining based on the published CDIP-ChatGLM3 visual-model framework	Provides post-symptomatic image evidence and disease category output	Supports grape symptom recognition; does not replace pre-symptomatic risk warning
LLM advisory	Fine-tuned ChatGLM3-based advisory model from the CDIP-ChatGLM3 framework	Provides bounded consultation, explanation, and agronomic advisory text	Knowledge-support function only; does not replace models, experts, labels, or regulations

2.6 Closed-loop workflow and evaluation boundaries

2.6.1 Risk-task-protection feedback logic

The closed-loop workflow starts from the daily disease-risk output generated for an active crop season. The platform translates model output into user-facing risk states, such as protected, low-risk, favourable, or high-risk periods. When weather, phenology, and disease-risk outputs indicate a favourable infection window, the crop-season page and notification interface display the corresponding risk state and prompt the user to review the field condition and treatment window.

The notification is linked to the crop season rather than to a stand-alone warning message. From the warning page, the user can enter the plant-protection workflow, inspect the suggested treatment timing, and create a plant-protection task if intervention is required. This design connects risk viewing, decision review, and task creation within the same field-season context.

After a plant-protection task is created, the user can keep it pending, complete it after field operation, or leave it unexecuted. A completed task records the execution date and pesticide or fungicide information and is then used by the platform as management feedback. In the current workflow, a completed fungicide-related task activates a protection period or low-risk management state. During this period, the crop-season risk display is updated to a protected or low-risk state rather than continuing to show the original infection-risk status without considering the completed intervention.

When the protection period ends, the platform resumes dynamic risk prediction using the updated weather, phenology, disease-risk input, and management history. If a new favourable or high-risk window appears, the system again displays the risk state and prompts the user to review the need for a new plant-protection task. The resulting cycle is: risk prediction, risk-state display, user notification, task creation, task completion, protected-state update, and renewed risk prediction.

The image-recognition and LLM-assisted advisory modules provide an additional entry point when users observe symptoms during the crop season. A user can upload a field image for disease identification, and the platform returns a grape-disease diagnosis result that can be reviewed together with the current crop-season risk state. For downy mildew, this image-based confirmation can be aligned with the risk-prediction workflow. For other supported grape diseases, such as powdery mildew, the module provides diagnostic support without being treated as an equivalent risk-forecasting workflow.

The LLM-assisted function is positioned as a supplementary advisory and knowledge-support channel. Within the field-season context, it supports agronomic consultation, interpretation of standard management guidance, and community-style discussion around routine grape production practices. Risk forecasting and image-based disease diagnosis remain handled by the dedicated analytical modules; LLM outputs are treated as auxiliary information for learning, communication, and management planning rather than as a replacement for validated models, expert diagnosis, pesticide labels, or local regulations.

2.6.2 Backend records, crop-season replay, and evaluation boundaries

Backend database exports from GrapeMaster were used to check whether the platform had generated the record types required by the proposed workflow. The exported tables included user and profile tables, field records, crop-season records, notifications, plant-protection tasks, irrigation tasks, fungicide and pesticide records, weather-related records, phenology records, disease-risk records, image-recognition and advisory records, disease-report records, note records, and message records. These tables were treated as internal operational records for workflow verification.

Account-level screening was applied before summarizing GrapeMaster records. User records were first integrated by user identifier, custom-user identifier, and phone-number-type account fields. Accounts with no business records across field, crop-season, notification, task, fungicide/pesticide, AI/message, forum, disease-report, and note tables were excluded. Accounts without both field records and crop-season records were also excluded because they could not support a crop-season-centered workflow. A single account located only in Linwei District was excluded according to the data-screening inventory. After screening, 47 accounts remained from 95 deduplicated accounts and were used as the candidate account set for describing GrapeMaster operational-record coverage.

The screened records were used for four workflow checks: field and crop-season creation and linkage; attachment of weather, phenology, risk, and notification records to crop seasons; creation and status updating of plant-protection tasks and pesticide/fungicide records; and linkage of image, advisory, disease-report, and note records to management contexts.

The study uses different evidence sources for different claims. The screened GrapeMaster backend records and workflow outputs are used to verify whether the platform has implemented field creation, crop-season management, weather and phenology records, disease-risk output, notification delivery, plant-protection task handling, fungicide-feedback logic, image upload, LLM consultation, and management-record storage. These records support platform feasibility, data-linkage coverage, and workflow validation.

Module evidence is used to support the trustworthiness of the analytical configuration in GrapeMaster. The downy mildew initial-infection model contributes calibrated and validated pre-symptomatic disease-risk capability. The phenology module contributes crop-stage timing required for risk interpretation and warning windows. The image-recognition module contributes post-symptomatic grape disease evidence, including downy mildew-related diagnosis where applicable and auxiliary recognition of other grape diseases such as powdery mildew. The LLM module contributes bounded knowledge explanation and interaction. These modules are evaluated as parts of the platform workflow, not as independent standalone claims in this manuscript.

The present study does not use GrapeMaster backend records to claim real-world user adoption, confirmed pesticide reduction, or economic benefits. Such claims require future production-scale deployment, controlled comparison with grower practice, quantitative treatment records, disease outcomes, economic accounting, and user-centered evaluation.

3 Results

3.1 Deployed platform architecture and operational data coverage

The deployed GrapeMaster backend produced operational records across the main objects required by the crop-season-centered workflow. Account-level screening was applied before summarizing these records. The raw database export contained 95 deduplicated registered accounts; 47 accounts were retained because they contained both field records and crop-season records, whereas 48 accounts without this minimum operational linkage were excluded. The resulting statistics therefore describe operational data coverage rather than raw registration volume.

The retained records covered the core data objects needed to connect vineyard setup, environmental state, analytical output, field operations, symptom evidence, and advisory interaction. The cleaned backend export contained 139 vineyard fields, 128 crop seasons, 126 weather payloads, 126 phenology payloads, and 252 disease-risk or model request-response payloads linked to retained crop seasons. It also contained 3590 notifications, 151 plant-protection or irrigation tasks, and 234 fungicide or pesticide records. These records show that the platform generated the database objects required for risk delivery, task execution, and protection feedback rather than only storing static field information.

Field evidence and consultation records were also present in the retained account set. The backend contained 94 field-note or disease-report records, 2808 image-analysis records, 1360 advisory or message records, and 27 forum records after screening. These records provide evidence that post-symptomatic observations, image-based assistance, and advisory interactions can be stored alongside field, crop-season, risk, and task data. Figure 3 summarizes this backend linkage, while Table 3 reports the corresponding record coverage. Together, these results show that the deployed platform produced the data objects needed for the subsequent mobile workflow and closed-loop result sections.

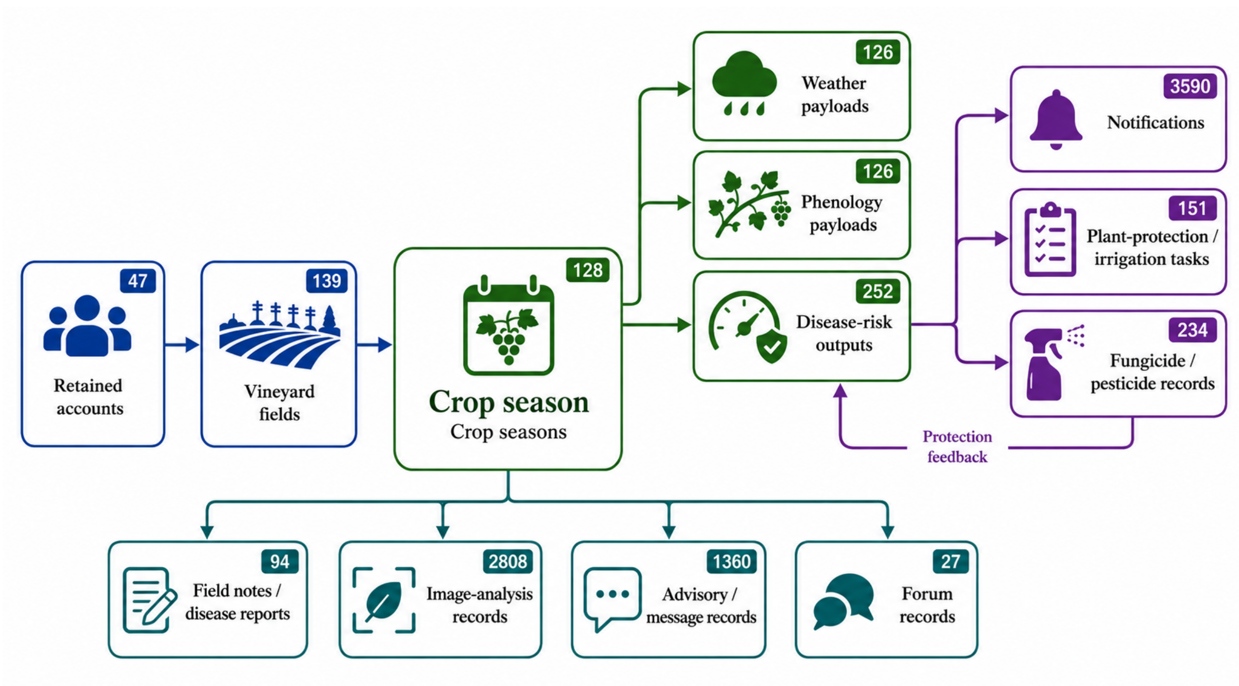


Figure 3. Operational backend data coverage and crop-season-centered linkage in GrapeMaster. After excluding accounts without field and crop-season records, the retained backend records covered account and field setup, crop-season objects, weather and phenology payloads, disease-risk outputs, notifications, field tasks, fungicide or pesticide records, field evidence, image-analysis records, advisory messages, and forum records. The crop season served as the central data anchor linking analytical state, risk delivery, field operations, protection feedback, and observation records.

Table 3. Audit-oriented summary of cleaned backend records used to support the platform-coverage result.

Audit focus	Cleaned statistic	Linkage criterion	Result meaning
Account screening	47 of 95 deduplicated accounts retained	At least one field and one crop season	Excludes inactive registration-only accounts
Field-season foundation	139 fields and 128 crop seasons	Retained account, field UUID, and crop-season UUID	Defines the operational denominator
Analytical payload completeness	126 of 128 crop seasons with weather, phenology, risk, and request-response payloads	Shared crop-season UUID	Shows analytical-state generation for most retained crop seasons
Risk-to-operation linkage	3590 notifications and 151 field-operation tasks	Retained users or crop seasons	Shows delivery into user-facing operations
Protection-feedback availability	234 fungicide or pesticide records linked to 141 plant-protection tasks	Task UUID	Shows protection inputs available for feedback
Observation and consultation archive	94 field-note or disease-report records, 2808 image-analysis records, and 1360 advisory or message records	Retained accounts, fields, crop seasons, or uploaded images	Shows symptom and consultation evidence retained with operational data

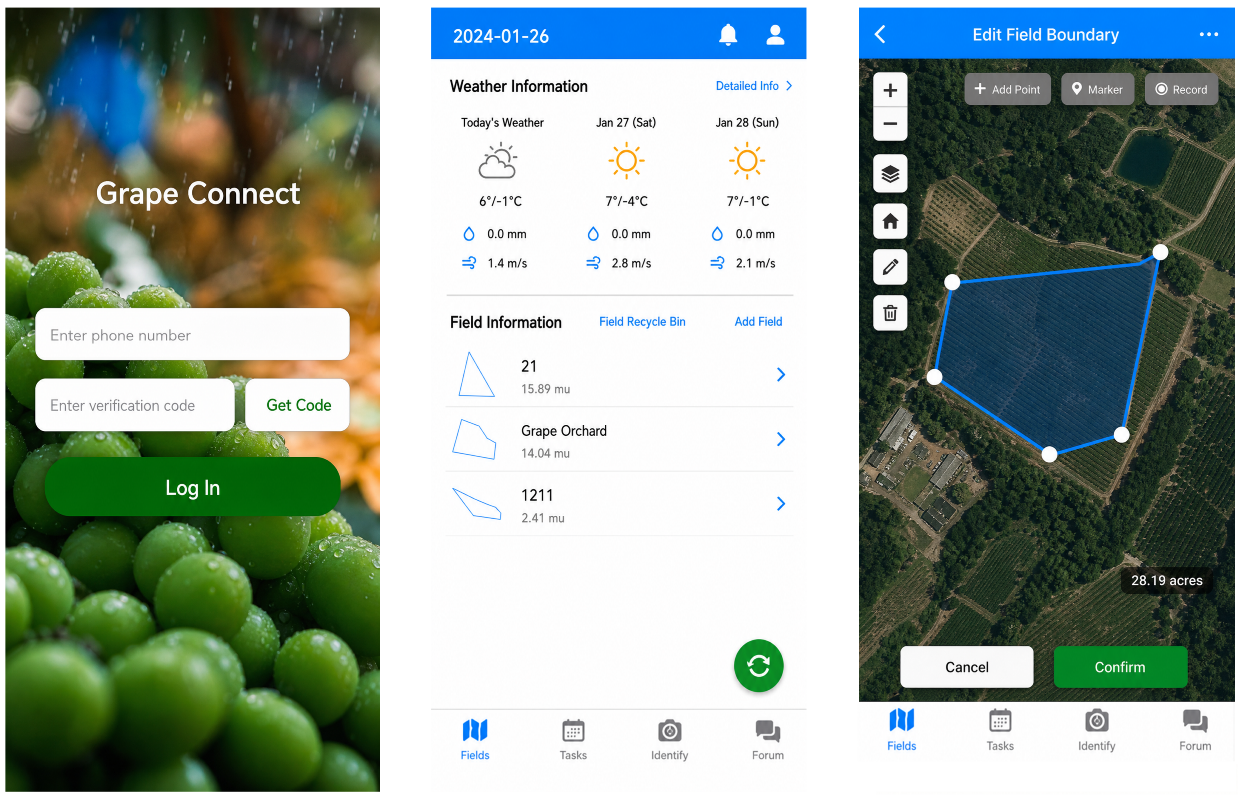
3.2 Crop-season-centered data linkage and mobile workflow implementation

The mobile client implemented the field and crop season as the operational entry points for vineyard management. Field-management interfaces allowed users to create, inspect, select, share, delete, recover, and revisit vineyard fields. Each field card provided access to crop-season functions, including weather, phenology, disease-risk warning, notification, task, note, and image-related records. This result shows that the platform organized user interaction around field-season objects rather than presenting weather, warning, task, and image functions as separate tools.

Weather and phenology outputs were displayed within the same crop-season context. The weather interface provided near-term weather information for the selected field, while the phenology interface presented crop-stage status generated by the platform phenology workflow. An important implementation result is that phenology was not treated as a manually entered external descriptor. It was displayed as a model-derived crop-season state that could be viewed together with local weather and later used to interpret disease-risk timing and task windows.

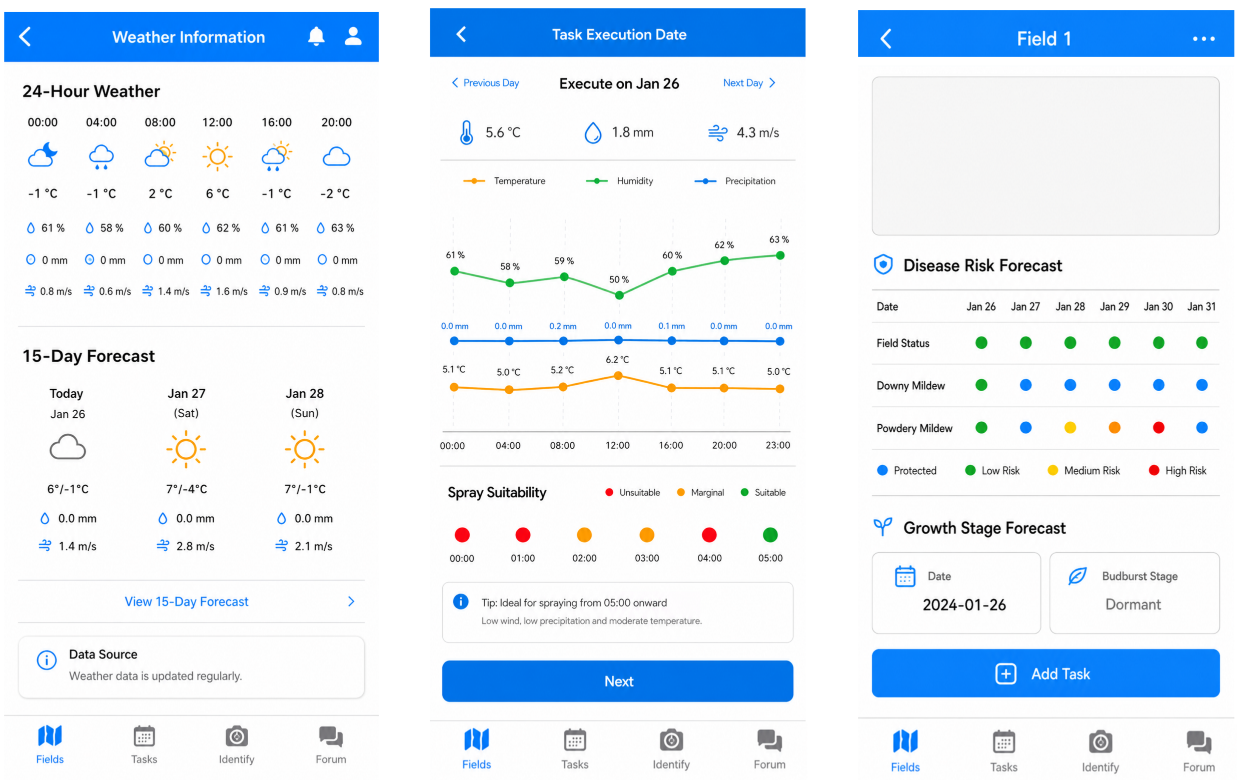
3.3 Trustworthy analytical modules embedded in GrapeMaster

The analytical configuration of GrapeMaster was supported by module-specific evidence rather than by backend records alone. FSIM-S provided the calibrated downy mildew first-infection risk engine for pre-symptomatic warning in the Guangxi humid subtropical vineyard context. The phenology



(a) **(b)** **(c)**

Figure 4. Mobile field-management interfaces in GrapeMaster. The field workspace provides the operational entry point for field creation, field inspection, crop-season access, field sharing, deletion, recovery, and linked management functions.



(a) **(b)** **(c)**

Figure 5. Mobile weather and phenology interfaces in GrapeMaster. Field-specific weather information and model-derived crop-stage status are displayed within the same crop-season workspace, supporting risk interpretation and management timing.

module provided crop-stage timing for crop-season interpretation and disease-risk windows. The grape disease-recognition module provided post-symptomatic visual evidence after grape-focused retraining. The LLM advisory function provided bounded consultation and management-oriented explanation as a knowledge-support channel.

Together, these modules supplied the analytical basis for the platform workflow. The disease-risk and phenology modules supported pre-symptomatic and crop-stage-aware interpretation, while image recognition and LLM-assisted consultation supported symptom evidence and advisory records after field observations. The result is a configured platform in which analytical outputs are assigned distinct workflow roles instead of being treated as interchangeable predictions.

3.4 Operational delivery of disease-risk information

Disease-risk outputs were converted into a mobile timeline format for crop-season review. The timeline separated downy mildew risk, powdery mildew risk, and field-level integrated risk, allowing users to inspect how risk states changed through the season. The displayed states included non-seasonal or unfavourable periods, favourable or higher-risk periods, and protected periods when fungicide-related management records were present. This result is important because it shows that daily model outputs were not left as backend values; they were transformed into a date-aligned sequence that could be read as management information.

The warning and notification interfaces provided the next step from risk display to user action. Risk-related and crop-season-related events could be delivered as notifications, reviewed by the user, and used as entry points for related field operations. The notification record therefore acted as a bridge between analytical output and field response. It kept warning information attached to the field-season context and made the event available for later review rather than presenting it as a transient message.

3.5 Risk-to-task execution, protection feedback, and symptom-evidence functions

The platform implemented a user-facing route from risk information to field operation. Plant-protection tasks stored task type, execution date, task status, sprayer or container information, pes-

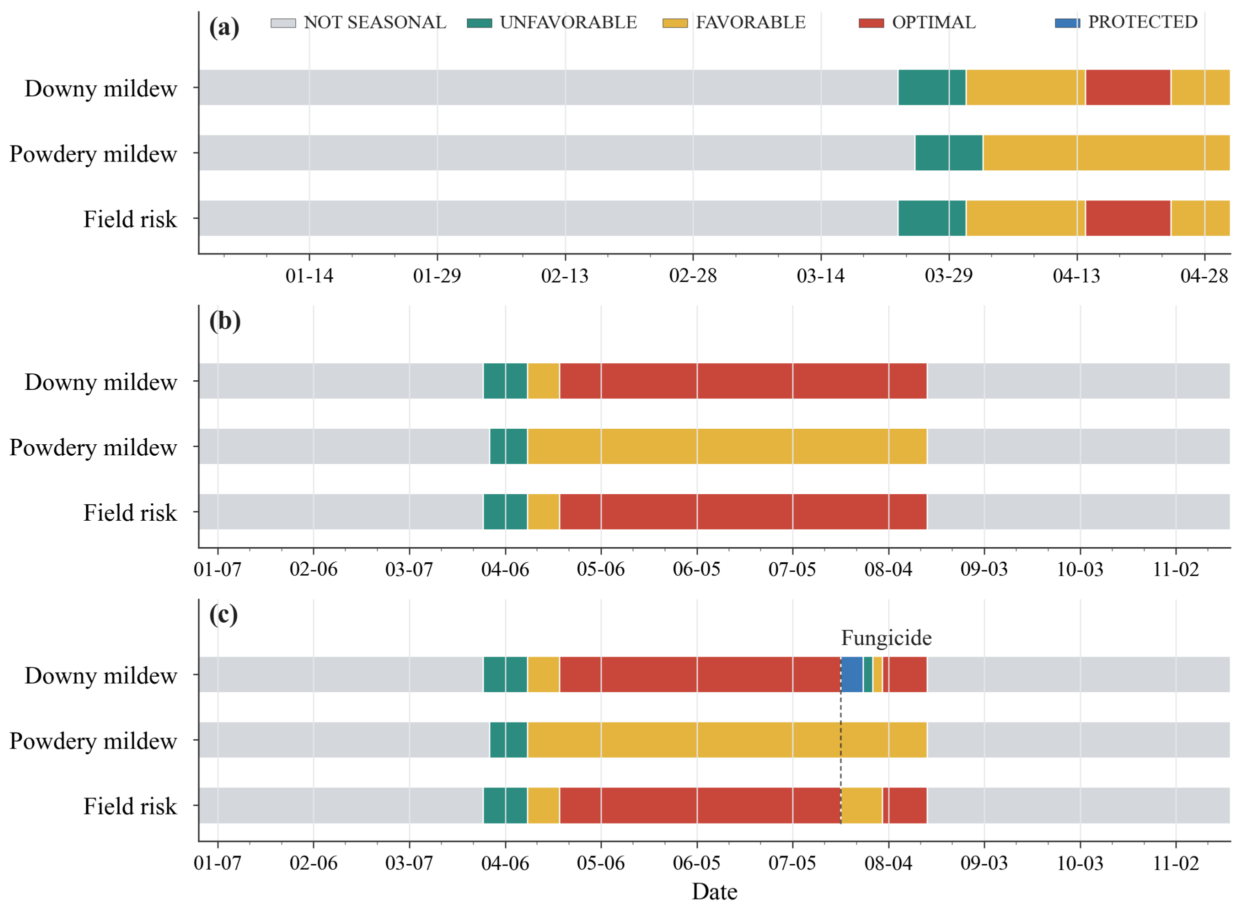
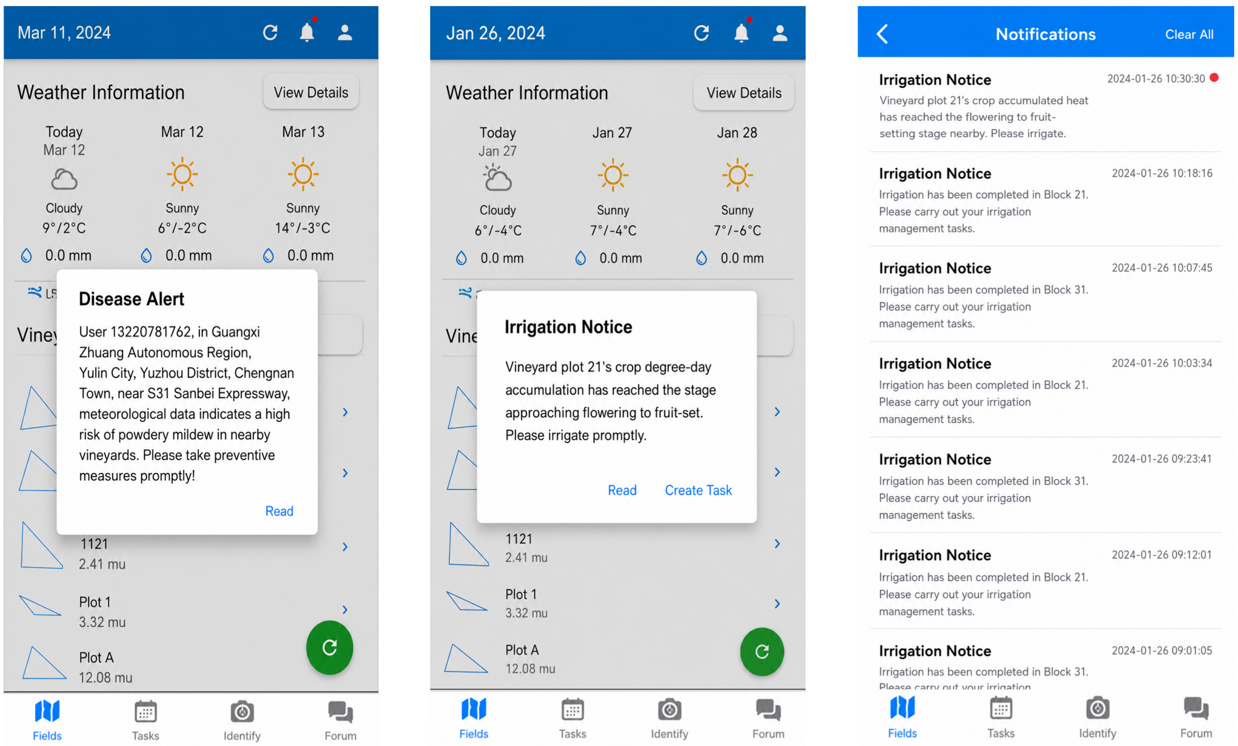


Figure 6. Disease-risk timeline displayed in the GrapeMaster mobile workflow. The timeline converts date-aligned model outputs into user-facing downy mildew risk, powdery mildew risk, and field-risk states, including protected periods when management records modify subsequent risk interpretation.



(a) (b) (c)

Figure 7. Alert and notification interfaces used to deliver crop-season events in GrapeMaster. Disease-risk and management-related events are presented as reviewable mobile notifications that remain linked to the field-season workflow.

ticide or fungicide records, and completion state. Completed fungicide-related tasks supplied the management feedback required to update protection status in subsequent disease-risk interpretation. This result shows that task records were not only calendar entries; they formed the operational feedback layer needed for risk-state updating.

Image-based disease assistance added a post-symptomatic evidence channel to the same crop-season workflow. Users could upload field images, receive a grape disease-recognition result, and continue with advisory interaction or disease reporting. These outputs were positioned as auxiliary symptom evidence and consultation records. They complement weather-driven warning and task feedback without replacing the disease-risk engine or expert and regulatory judgement.

3.6 Representative crop-season replay of the closed loop

Representative crop-season replay confirmed that the platform could connect the main workflow objects in sequence. A risk period could be represented in the disease-risk timeline, delivered through notification, converted into a plant-protection task, updated after task completion, reflected as a protection or low-risk state, and archived together with image, advisory, disease-report, or note records when available. The resulting record chain was: field and crop season, weather and phenology, disease-risk state, notification, task, fungicide or pesticide record, protection-state update, symptom evidence, advisory message, and management archive.

This replay result supports the core platform claim of the study. GrapeMaster was able to transform disease-risk information into executable and traceable workflow objects within a single crop-season context. The result does not by itself quantify production-scale pesticide reduction or long-term user adoption, but it demonstrates that the data objects and mobile workflow needed for such evaluation are present in the deployed platform.

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